

- e)  $p \leftrightarrow q$  —————→ I bought a lottery ticket this week if and only if I won the million dollar jackpot.
- f)  $\neg p \rightarrow \neg q$  —————→ If I did not buy a lottery ticket this week, then I did not win the million dollar jackpot.
- g)  $\neg p \wedge \neg q$  —————→ I did not buy a lottery ticket this week and I did not win the million dollar jackpot.
- h)  $\neg p \vee (p \wedge q)$  —————→ <sup>Either</sup> I did not buy a lottery ticket this week, or, I bought a lottery ticket this week and I also won the million dollar jackpot.

11) Let  $p$  and  $q$  be the propositions "Swimming at the New Jersey shore is allowed" and "Sharks have been spotted near the shore", respectively. Express each of these compound propositions as an English sentence.

- a)  $\neg p$  —————→ Swimming at the New Jersey shore is not allowed.
- b)  $p \wedge q$  —————→ Swimming at the New Jersey shore is allowed and Sharks have been spotted near the shore.
- c)  $\neg p \vee q$  —————→ Swimming at the New Jersey shore is not allowed or Sharks have been spotted near the shore.
- d)  $\neg q$  —————→ Sharks have not been spotted near the shore.
- e)  $p \rightarrow \neg q$  —————→ If Swimming at the New Jersey shore is allowed, then sharks have not been spotted near the shore.
- f)  $\neg q \rightarrow \neg p$  —————→ If Sharks have not been spotted near the shore, then swimming at the New Jersey shore is allowed.
- g)  $\neg p \rightarrow \neg q$  —————→ If Swimming at the New Jersey shore is not allowed, then sharks have not been spotted near the shore.
- h)  $p \leftrightarrow \neg q$  —————→ Swimming at the New Jersey shore is allowed if and only if Sharks have not been spotted near the shore.